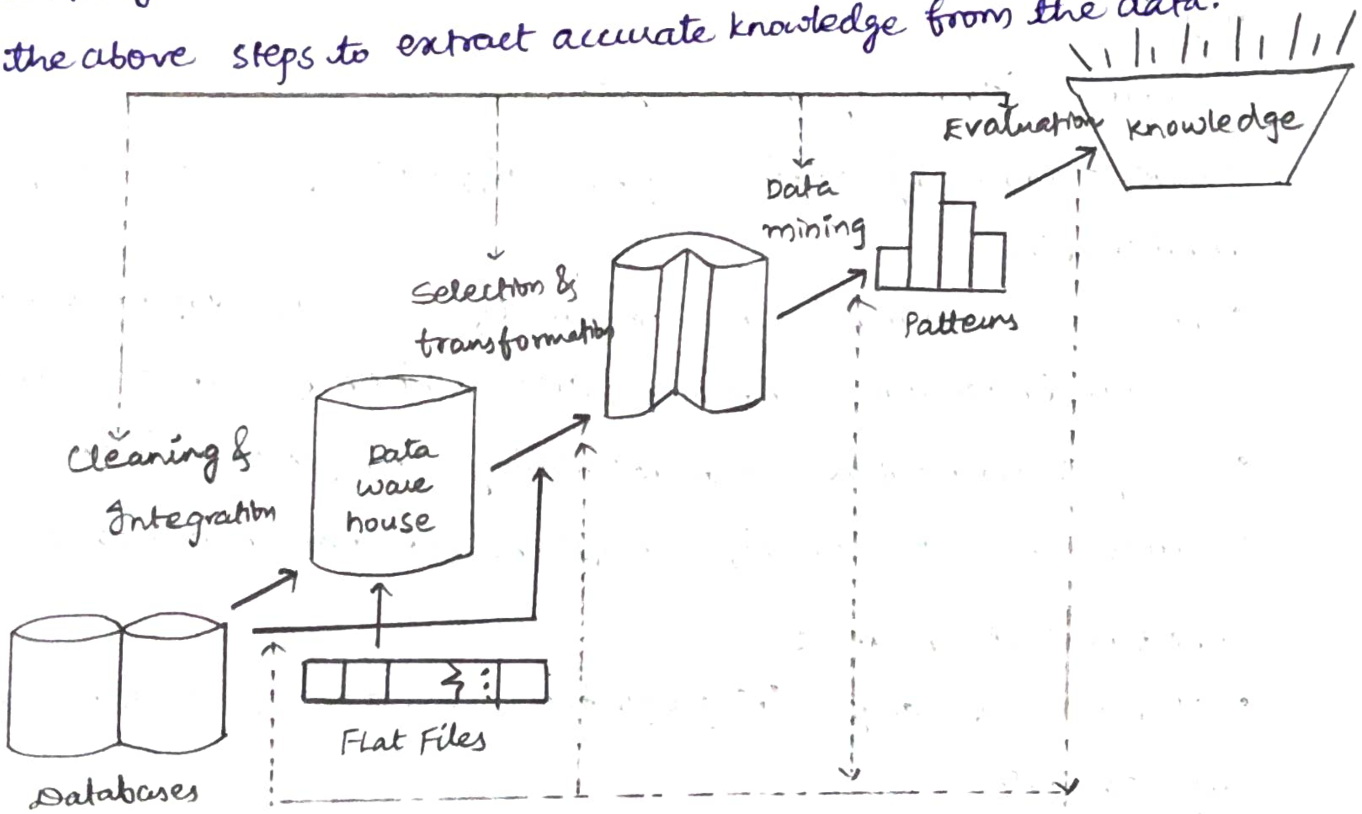
1.a. What is data mining? Explain the various steps involved in the Knowledge Discovery process with neat sketch.

Ans: The need of data mining is to extract useful information from large datasets and use it make predictions or better decision-making. Nowadays, datamining is used in almost all places where a large amount of data is stored and processed.

For example: Banking Sector, Market Basket Analysis.

Data Mining is also known as Knowledge Discovery from Data or KDD. The KDD process is an iterative process and it requires multiple iterations of the above steps to extract accurate knowledge from the data.



Data cleaning: Removes noisy, inconsistent, and irrelevant data. It includes handling missing values and removing errors.

Data Integration: Combines data from multiple sources into a unified dataset.

Data Selection: Selects only the relevant data required for analysis and removes unnecessary data.

Data Transformation: Converts data into a suitable format for mining using techniques such as normalization, aggregation, and discretization.

Data Mining: Applies algorithms to discover useful patterns, relationships, and trends in the data. Techniques include classification, clustering, and association rule mining.

Pattern Evaluation: Evaluates the discovered patterns based on their - accuracy, usefulness, and relevance.

Knowledge Representation: Represents the discovered knowledge in an understandable from such as reports, graphs, or visualizations.

1.6. Describe the major issues in data mining.

Ans: The major issues in data mining are mainly divided into three categories:

1. Mining Methodology & User Interaction:

- Diverse Knowledge Mining: Supports different types of knowledge & user requirements.
- Interactive mining: Allows users to refine queries & explore results.
- Results Presentation: Presents patterns in simple & understandable formats.
- Noisy / Incomple Data: Handles missing & noisy data to improve accuracy.
- Pattern Evaluation: Identifies useful, novel & non-redundant patterns.

2. Performance Issues:

- Algorithm Efficiency: Mining algorithms should be efficient & scalable for large datasets.
- Parallel & Distributed Mining: Processes large datasets using multiple systems simultaneously.
- Incremental Mining: Updates existing patterns without repeating the entire mining process.

3. Diverse Data Types Issues:

- Complex Data: Handles relational, spatial, temporal, and multimedia data.

- Heterogeneous Sources: Integrates data from different structured, semi-structured, and unstructured sources.

2. What are the kinds of data that can be mined? Explain them briefly.

Ans: The following are the most basic forms of data for mining

i) Database Data (Relational Database):

A database systems is a collection of related data & software programs to manage & access that data.

A relational database stores data in tables. Each table has a unique name, attributes, and records. Every record represents an object & is identified by a unique way.

column

Eg:

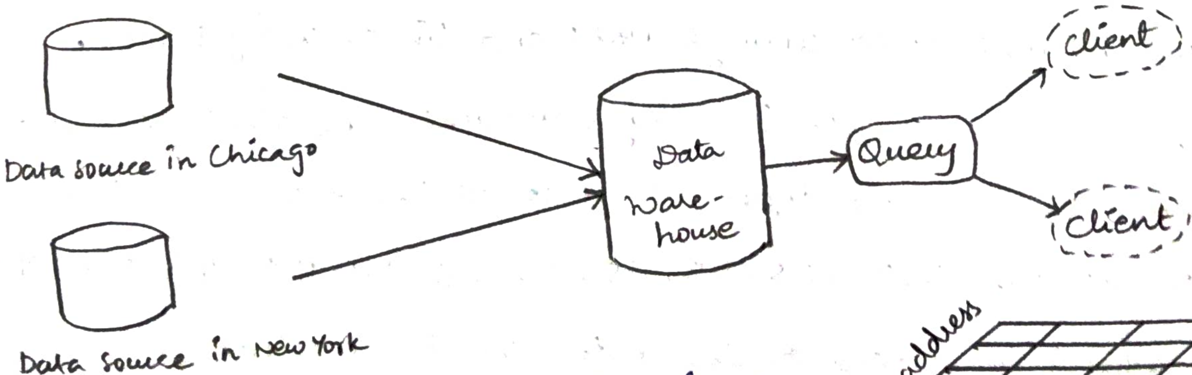
field →

row →

Name	age	Gender
Akhil	25	Male
Sai	25	Male
Varsha	28	Female
Bindu	20	Female

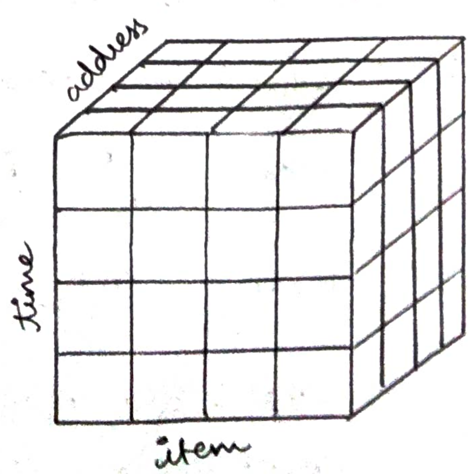
ii) Data ware house Data:

It is a repository of information collected from multiple Sources, Stored under a unified schema, and usually residing at a single site.



A data warehouse uses a multidimensional model called data cube.

- Each dimension represents an attribute.
- Each cell contains summarized data.



iv) Transactional Data:

A Transactional database stores information about transactions - such as purchases, bookings, or clicks - organized by time.

Each transaction has: Transaction ID and list of items

It follows ACID properties to ensures reliable processing.

Eg:

TID	List of items
T1	A, C, E
T2	B, C, D, E, F
T3	A, C, E
T4	B, E, F

v) Multimedia Data:

Multimedia Databases stores different types of media such as image, videos, audio, and text. They handle multiple file formats like .jpg, .mp3, .mp4, .swf, .txt.

vi) Spatial database:

A spatial database stores & manages spatial or geographic data such as locations & features. Data is represented using points, lines, polygons.

vii) World wide web:

The WWW is a collection of web pages & resources such as text, audio, and video, accessed through URLs and linked using HTML.

3.* Suppose that the data for analysis includes the attribute age. The age values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Ans: i) what is the mean of the data? what is median?

$$\text{Mean} = \frac{\Sigma x}{n} = \frac{809}{27} = 29.96$$

$$\text{Median} = \frac{n+1}{2} = \frac{28}{2} = 14^{th} \text{ value is } 25.$$

ii) what is the mode of the data? Comment on the data modality.

Count the repeated values: 25 occurs 4 times & 35 occurs 4 times

Therefore, Mode = 25, 35

Since there are two modes, the data is bimodal.

iii) Find the first Quartile (Q1) and third Quartile (Q3).

For $n = 27$: Q_1 position $= \frac{n+1}{4} = \frac{28}{4} = 7$

The 7^{th} observation $= 20$. $\Rightarrow Q_1 = 20$

Q_3 position $= \frac{3(n+1)}{4} = \frac{(28)3}{4} = 21$

The 21^{st} observation $= 35 \Rightarrow Q_3 = 35$

iv) Show the histogram of the data.

4.a. Why are preprocess the data? What are the major tasks in data - preprocessing? Explain them.

Ans: Data Preprocessing :- It is the process of converting raw data into a clean, consistent, and suitable format before it is used for data - mining or analysis.

Why do we preprocess data?

Improve data quality, Handle missing values, Remove noise, Make data consistent, Improve Efficiency. Reduce redundancy,

Major Tasks are:

i) Data Cleaning: It removes or corrects incorrect, incomplete, noisy, or inconsistent data. Common techniques include: Handling missing values, Remove duplicating records, Smoothing noisy data, etc..

ii> Data Integration: It combines data from multiple sources into a single data store. Problems addressed here are, Duplicate records, Conflicting values, Different data formats.

iii> Data Reduction: It reduces the volume of data while preserving important information. This makes data mining faster and more effi-cient. It includes dimensionality reduction, Numerosity reduction and Data compression.

iv> Data Transformation: Data transformation converts data into a suitable form or format for data mining algorithms. Common techni-ques are Normalization, Aggregation, Generalization, Discretization

v> Data Discretization: It converts continuous numerical attributes into intervals or categories.

4.b. What is data cleaning? How to handle missing data and noisy data? Explain them.

Ans: <u>Data Cleaning</u>:

This process involves addressing missing values, smoothing noisy data, detecting and removing outliers, and resolving inconsistencies. It is essential for gaining trust in the data mining results.

<u>Handle missing values</u>:

- Ignore the Juple: useful when the class is missing but less effe-ctive for large datasets.
- File Manually: Effective for small datasets but impractical for large ones.
- Use global constant: Replace missing values with constants like "Unknown".
- Use Attribute Mean/Median: Replace missing values with the mean or median of the attribute values.

Handling Noisy Data:

Noise refers to random errors or variances in data. Common techniques are:

a. Binning: Divides sorted data into "bins" and smooths data based on nearby values.

Eg: Sorted Data: 4, 8, 15, 21, 21, 24, 25, 28, 34

Bin 1: 4, 8, 15

Bin 2: 21, 21, 24

Bin 3: 25, 28, 34

Mean smoothing: Bin 1 = 9, 9, 9

Median smoothing: Bin 1 = 8, 8, 8

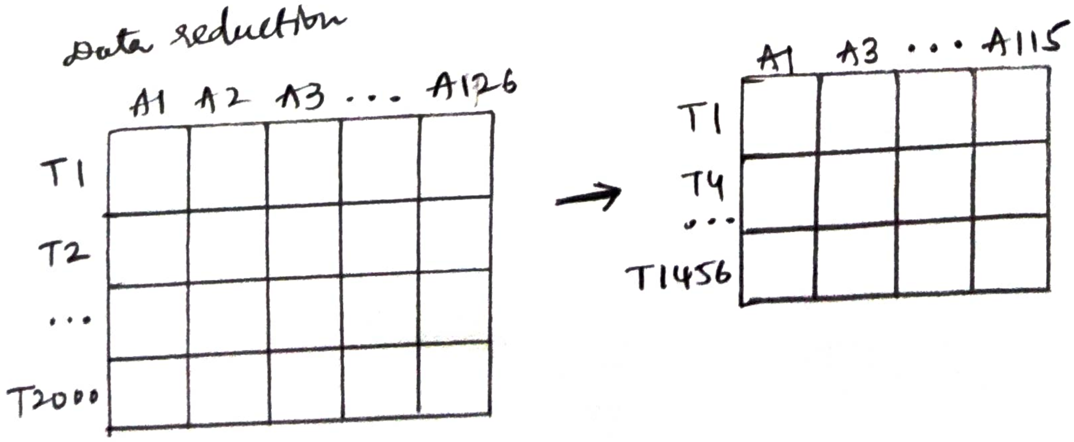
Boundary smoothing: Bin 1 = 4, 4, 15

b. Regression: It predicts numerical values by studying the relationship b/w variables. Linear regression uses a line, while multiple linear regression uses more than two variables.

c. Clustering: Groups similar data points into clusters, identifying outliers as values that fall outside these clusters.

5. Explain the various data reduction techniques.

Ans: Data reduction is the process of reducing the size of a dataset while keeping important information intact. It helps to make data mining faster & more efficient without losing accuracy.



Techniques:

- Data sampling: Select a subset of data that represents the whole dataset.
- Dimensionality Reduction: Reduce the number of features or combine related ones.
- Data Compression: Store data in a smaller from using techniques like lossless or lossy compressions.